Create a network with 2 routers running both EIGRP and static routes, configure 4 VLANs across switches, setup DHCP relay agents, add web and DNS servers in different VLANs, test connectivity and analyze DHCP and DNS packets.

OBJECTIVES

1. To understand the working principles of EIGRP, static routing, VLAN segmentation, and DHCP relay agents.

2. To configure these protocols and services on routers and switches using Cisco Packet Tracer to enable automatic IP assignment, route discovery, and communication between different VLANs and servers.

THEORY

1. EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced distance-vector protocol (often called hybrid) developed by Cisco.

Operation: It uses the Diffusing Update Algorithm (DUAL) to calculate the shortest path. It sends partial updates only when topology changes occur, rather than full periodic updates.

Application: Used in Cisco-based enterprise networks requiring fast convergence and efficient bandwidth usage.

2. Static Routing

Static routing is the manual configuration of network routes by a network administrator.

Operation: It works by explicitly defining the path that a packet must take to reach a specific destination network. It does not automatically update if the network topology changes.

Application: Used in small networks or for specific routing overrides where simplicity, security, and low router overhead are required.

3. VLAN (Virtual Local Area Network)

VLANs logically segment a physical network into multiple isolated broadcast domains.

Operation: Switches assign physical ports to specific VLAN IDs. To allow communication between these isolated networks, a router with sub-interfaces (Router-on-a-Stick) is used.

Application: Used to improve network security, manage traffic efficiently, and reduce unnecessary broadcast traffic.

4. DHCP Relay Agent

A DHCP relay agent forwards DHCP requests from clients in one subnet to a DHCP server located in a different subnet.

Operation: It listens for DHCP broadcast messages from clients and converts them into unicast packets, forwarding them to the specified IP address of the DHCP server using the `ip helper-address` command.

Application: Used in enterprise networks to centralize DHCP management rather than deploying a separate DHCP server for every VLAN.

5. DNS and Web Servers

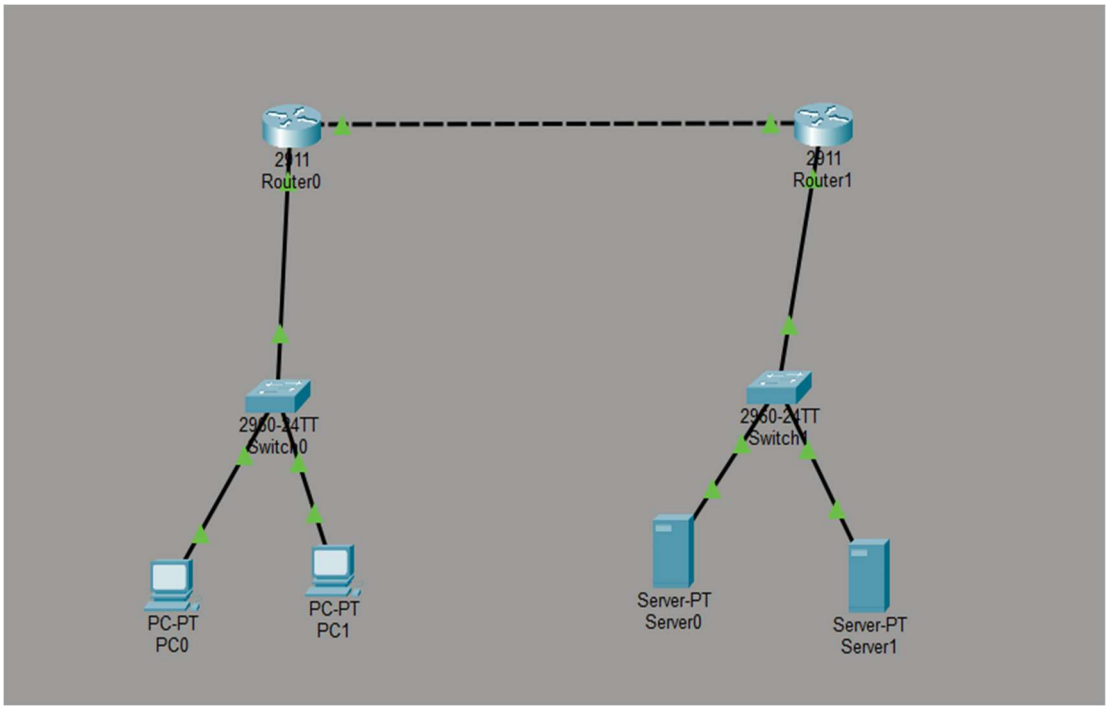
DNS (Domain Name System) translates human-readable domain names into IP addresses, while Web servers host website content.

Operation: DNS operates primarily on UDP port 53 to resolve queries. Web servers use TCP port 80 (HTTP) or 443 (HTTPS) to serve web pages.

Application: Used to provide user-friendly access to network resources and host intranet/internet applications.

NETWORK TOPOLOGY

A topology was created using Router0 and Router1 connected via a serial/WAN link. Switch0 connects PC0 and PC1 to Router0. Switch1 connects Server0 (Web Server) and Server1 (DNS Server) to Router1. The network is divided into 4 VLANs: VLAN 10 and VLAN 20 on Switch0, and VLAN 30 and VLAN 40 on Switch1.



CONFIGURATION TABLE

IP Configuration Table

Device	Interface/VLAN	IP Address	Subnet Mask	Gateway
Router0	Gi0/0 (to SW0)	192.168.1.1	255.255.255.0	—
Router0	Gi0/1 (to R1)	10.0.0.1	255.255.255.252	—
Router1	Gi0/0 (to SW1)	192.168.2.1	255.255.255.0	—
Router1	Gi0/1 (to R0)	10.0.0.2	255.255.255.252	—
VLAN 10	SVI on R0	172.16.10.1	255.255.255.0	—
VLAN 20	SVI on R0	172.16.20.1	255.255.255.0	—
VLAN 30	SVI on R1	172.16.30.1	255.255.255.0	—
VLAN 40	SVI on R1	172.16.40.1	255.255.255.0	—
Web Server	VLAN 30	172.16.30.100	255.255.255.0	172.16.30.1
DNS Server	VLAN 40	172.16.40.100	255.255.255.0	172.16.40.1
PC0 (VLAN10)	VLAN 10	DHCP	DHCP	172.16.10.1
PC1 (VLAN20)	VLAN 20	DHCP	DHCP	172.16.20.1

COMMANDS USED

Switch switch0 — VLANs 10 and 20

```
SWITCH> enable SWITCH# configure terminal

SWITCH(config)# vlan 10 SWITCH(config-vlan)# name VLAN10_USERS

 SWITCH(config-vlan)# exit SWITCH(config)# vlan 20

SWITCH(config-vlan)# name VLAN20_MGMT

SWITCH(config-vlan)# exit

! Assign PC1 port to VLAN 10

 SWITCH(config)# interface FastEthernet0/2

SWITCH(config-if)# switchport mode access

SWITCH(config-if)# switchport access vlan 10

SWITCH(config-if)# exit

! Assign PC2 port to VLAN 20

SWITCH(config)# interface FastEthernet0/3

SWITCH(config-if)# switchport mode access
```

```
SWITCH(config-if)# switchport access vlan 20
SWITCH(config-if)# exit
! Configure trunk port toward Router0
SWITCH(config)# interface FastEthernet0/1
SWITCH(config-if)# switchport mode trunk
SWITCH(config-if)# switchport trunk allowed vlan 10,20
SWITCH(config-if)# exit
SWITCH(config)# end
SWITCH# write memory
```

Switch switch1 — VLANs 30 and 40

```
SWITCH> enable
SWITCH# configure terminal
SWITCH(config)# vlan 30
SWITCH(config-vlan)# name VLAN30_SERVERS
SWITCH(config-vlan)# exit
SWITCH(config)# vlan 40
SWITCH(config-vlan)# name VLAN40_DNS
SWITCH(config-vlan)# exit
! Assign Web Server port to VLAN 30
SWITCH(config)# interface FastEthernet0/2
SWITCH(config-if)# switchport mode access
SWITCH(config-if)# switchport access vlan 30
SWITCH(config-if)# exit
! Assign DNS Server port to VLAN 40
SWITCH(config)# interface FastEthernet0/3
SWITCH(config-if)# switchport mode access
SWITCH(config-if)# switchport access vlan 40
SWITCH(config-if)# exit
```

```
! Configure trunk port toward Router1
SWITCH(config)# interface FastEthernet0/1
SWITCH(config-if)# switchport mode trunk
SWITCH(config-if)# switchport trunk allowed vlan 30,40
SWITCH(config-if)# exit
SWITCH(config)# end
SWITCH# write memory
```

Router0 — Sub-Interfaces for VLAN 10 & 20

```
Router0> enable
Router0# configure terminal
! Physical interface toward SW0 (must be up, no IP on main)
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# no shutdown
Router0(config-if)# exit
! Sub-interface for VLAN 10
Router0(config)# interface GigabitEthernet0/0.10
Router0(config-subif)# encapsulation dot1Q 10
Router0(config-subif)# ip address 172.16.10.1 255.255.255.0
Router0(config-subif)# exit
! Sub-interface for VLAN 20
Router0(config)# interface GigabitEthernet0/0.20
Router0(config-subif)# encapsulation dot1Q 20
Router0(config-subif)# ip address 172.16.20.1 255.255.255.0
Router0(config-subif)# exit
! WAN link to Router1
Router0(config)# interface GigabitEthernet0/1
Router0(config-if)# ip address 10.0.0.1 255.255.255.252
Router0(config-if)# no shutdown
```

```
Router0(config-if)# exit
Router0(config)# end
Router0# write memory
```

Router1 — Sub-Interfaces for VLAN 30 & 40

```
Router1> enable
Router1# configure terminal
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# no shutdown
Router1(config-if)# exit
! Sub-interface for VLAN 30
Router1(config)# interface GigabitEthernet0/0.30
Router1(config-subif)# encapsulation dot1Q 30
Router1(config-subif)# ip address 172.16.30.1 255.255.255.0
Router1(config-subif)# exit
! Sub-interface for VLAN 40
Router1(config)# interface GigabitEthernet0/0.40
Router1(config-subif)# encapsulation dot1Q 40
Router1(config-subif)# ip address 172.16.40.1 255.255.255.0
Router1(config-subif)# exit
! WAN link to Router0
Router1(config)# interface GigabitEthernet0/1
Router1(config-if)# ip address 10.0.0.2 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# end
Router1# write memory
```

EIGRP on Router0

```
Router0# configure terminal
Router0(config)# router eigrp 1
Router0(config-router)# network 172.16.10.0 0.0.0.255
Router0(config-router)# network 172.16.20.0 0.0.0.255
Router0(config-router)# network 10.0.0.0 0.0.0.3
Router0(config-router)# no auto-summary
Router0(config-router)# end
```

EIGRP on Router1

```
Router1# configure terminal
Router1(config)# router eigrp 1
Router1(config-router)# network 172.16.30.0 0.0.0.255
Router1(config-router)# network 172.16.40.0 0.0.0.255
Router1(config-router)# network 10.0.0.0 0.0.0.3
Router1(config-router)# no auto-summary
Router1(config-router)# end
```

Static Routes on Router0

```
Router0# configure terminal
Router0(config)# ip route 172.16.30.0 255.255.255.0 10.0.0.2 200
! Static route to VLAN 40 via Router1
Router0(config)# ip route 172.16.40.0 255.255.255.0 10.0.0.2 200
Router0(config)# end
```

Static Routes on Router1

```
Router1# configure terminal
! Static route to VLAN 10 via Router0
Router1(config)# ip route 172.16.10.0 255.255.255.0 10.0.0.1 200
! Static route to VLAN 20 via Router0
```

```
Router1(config)# ip route 172.16.20.0 255.255.255.0 10.0.0.1 200
Router1(config)# end
```

DHCP Pools on Router0

```
Router0# configure terminal
! Exclude static addresses from DHCP assignment
Router0(config)# ip dhcp excluded-address 172.16.10.1 172.16.10.10
Router0(config)# ip dhcp excluded-address 172.16.20.1 172.16.20.10
! DHCP Pool for VLAN 10
Router0(config)# ip dhcp pool VLAN10_POOL
Router0(dhcp-config)# network 172.16.10.0 255.255.255.0
Router0(dhcp-config)# default-router 172.16.10.1
Router0(dhcp-config)# dns-server 172.16.40.100
Router0(dhcp-config)# exit
! DHCP Pool for VLAN 20
Router0(config)# ip dhcp pool VLAN20_POOL
Router0(dhcp-config)# network 172.16.20.0 255.255.255.0
Router0(dhcp-config)# default-router 172.16.20.1
Router0(dhcp-config)# dns-server 172.16.40.100
Router0(dhcp-config)# exit
Router0(config)# end
```

DHCP Relay on Router0 Sub-Interfaces

```
Router0# configure terminal
! Relay on VLAN 10 sub-interface (points to itself as server)
Router0(config)# interface GigabitEthernet0/0.10
Router0(config-subif)# ip helper-address 172.16.10.1
Router0(config-subif)# exit
! Relay on VLAN 20 sub-interface
```

```
Router0(config)# interface GigabitEthernet0/0.20
Router0(config-subif)# ip helper-address 172.16.20.1
Router0(config-subif)# exit
Router0(config)# end
```

DHCP Relay on Router1

```
Router1# configure terminal
! Forward DHCP requests from VLAN 30 to Router0's DHCP server
Router1(config)# interface GigabitEthernet0/0.30
Router1(config-subif)# ip helper-address 10.0.0.1
Router1(config-subif)# exit
! Forward DHCP requests from VLAN 40 to Router0's DHCP server
Router1(config)# interface GigabitEthernet0/0.40
Router1(config-subif)# ip helper-address 10.0.0.1
Router1(config-subif)# exit
Router1(config)# end
```

Configure the Web Server (VLAN 30)

Click on Server-Web > Desktop tab > IP Configuration. Set a static IP:

• IP Address: 172.16.30.100

• Subnet Mask: 255.255.255.0

• Default Gateway: 172.16.30.1

• DNS Server: 172.16.40.100 Then go to Services tab > HTTP.

Ensure HTTP service is ON.

Configure the DNS Server (VLAN 40)

Click on Server-DNS > Desktop tab > IP Configuration. Set a static IP:

• IP Address: 172.16.40.100

• Subnet Mask: 255.255.255.0

• Default Gateway: 172.16.40.1

Go to Services tab > DNS. Turn DNS Service ON and add a record:

• Name: www.lab.com

• Type: A Record

• Address: 172.16.30.100

• Click Add

PC0 — VLAN 10

Click PC1 > Desktop > IP Configuration. Select DHCP. Wait for the IP assignment. Expected range: 172.16.10.11 – 172.16.10.254.

PC1 — VLAN 20

Click PC2 > Desktop > IP Configuration. Select DHCP. Expected range: 172.16.20.11 – 172.16.20.254.

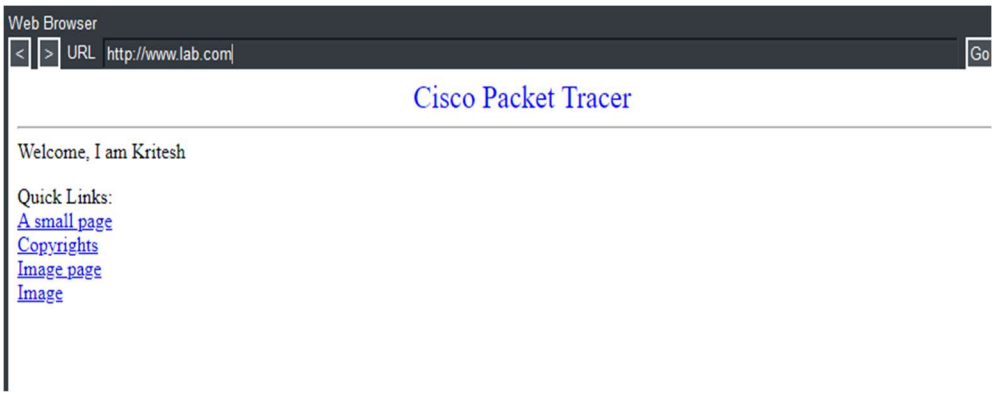
CONNECTION TEST

Ping Tests from PC0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.10.1

Pinging 172.16.10.1 with 32 bytes of data:

Reply from 172.16.10.1: bytes=32 time=15ms TTL=255
Reply from 172.16.10.1: bytes=32 time<1ms TTL=255
Reply from 172.16.10.1: bytes=32 time<1ms TTL=255
Reply from 172.16.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>
```

```
C:\>ping 172.16.40.100

Pinging 172.16.40.100 with 32 bytes of data:

Request timed out.
Reply from 172.16.40.100: bytes=32 time=11ms TTL=126
Reply from 172.16.40.100: bytes=32 time=11ms TTL=126
Reply from 172.16.40.100: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.40.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>
```

Web Browser Test



Router Verification

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.16.10.0/24 is directly connected, GigabitEthernet0/0.10
L       172.16.10.1/32 is directly connected, GigabitEthernet0/0.10
C       172.16.20.0/30 is directly connected, GigabitEthernet0/0.20
L       172.16.20.1/32 is directly connected, GigabitEthernet0/0.20
D       172.16.30.0/24 [90/28416] via 10.0.0.2, 00:44:52, GigabitEthernet0/1
D       172.16.40.0/24 [90/28416] via 10.0.0.2, 00:44:52, GigabitEthernet0/1

Router>
```

```
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)        Cnt Num
0   10.0.0.2        Gig0/1          13   00:46:42  40     1000  0   17
```

RESULT

The network was successfully configured using the various protocols and services. To verify the connectivity, a ping request was sent from PC0 as ping 172.16.10.1 (Gateway VLAN 10) similarly, to check the web configuration the https://www.lab.com was visited which was successfully achieved and different commands were tested for the router verification.

DISCUSSION AND CONCLUSION

For the completion of this project, I implemented the concept of dynamic routing alongside static routing to better understand how routers exchange information and handle manual route entries. EIGRP demonstrated faster convergence across the WAN link, while static routes allowed us to understand manual path definitions. Configuring 4 VLANs allowed us to understand logical network segmentation and Router-on-a-Stick inter-VLAN routing.

Additionally, setting up the DHCP relay agent introduced the concept of centralized IP management across different subnets, and analyzing the packets provided deep insight into how DHCP and DNS operate at the network layer. Hence the project was completed with a proper knowledge and implementation of advanced routing, VLANs, and server configurations using Cisco Packet Tracer.